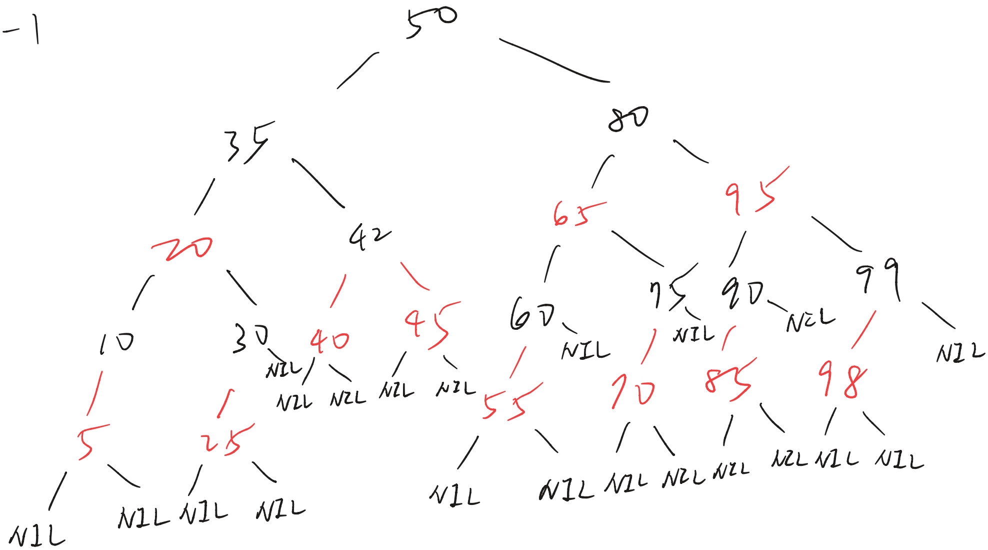
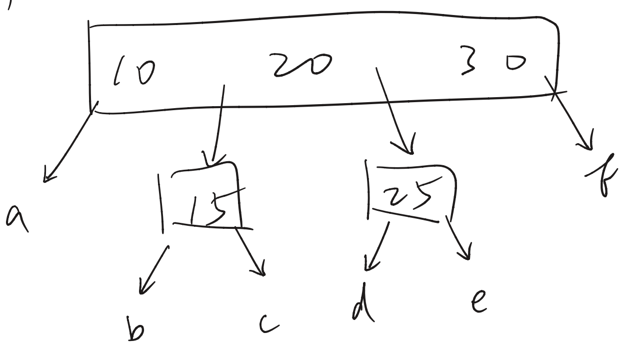
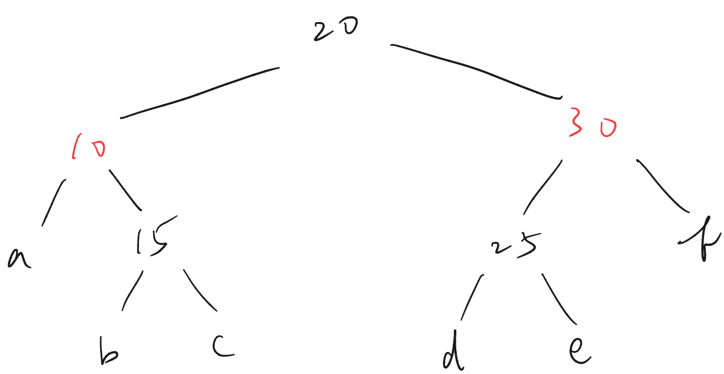
1-1


50
35
80
20
42
65
95
10
30
40
45
60
75
90
99
NIL
5
25
55
70
85
98
a

1-2(a)

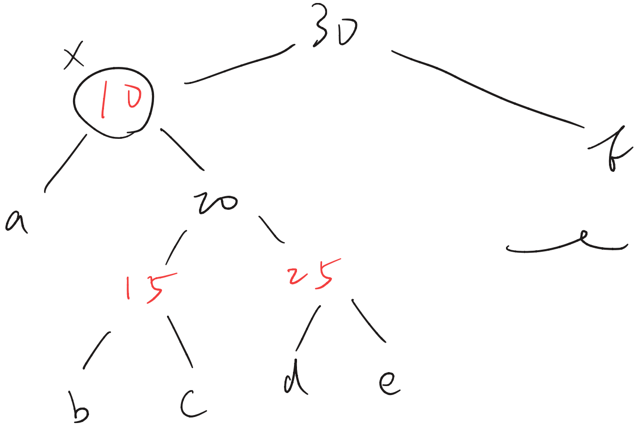

10
20
30
15
25
a
b
c
d
e
f

1-2(b)


20
10
30
a
15
25
f
b
c
d
e

1-2(c)

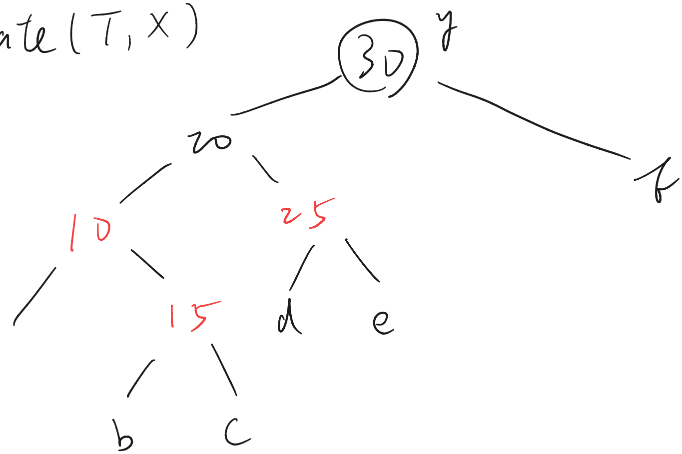
1-2(d)



rb tree of figure 1.2

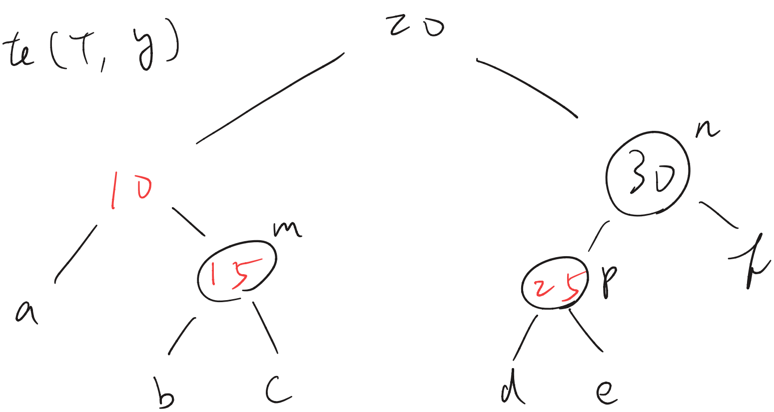
⇒ left-rotate(T, x)

⇒



⇒ right-rotate(T, y)

⇒



⇒ m.color = Black

n.color = Red

p.color = Black

1-5

the times it takes to read a disk block is

$1000 + 64t$ (microseconds)

the height of a tree with n nodes and each with max keys:

$$n = (2t-1)\sum_{i=1}^{h}(2t)^{i-1} = (2t-1)\left(\frac{(2t)^h - 1}{2t-1}\right) = (2t)^h - 1$$

$$\Rightarrow h = \log_{2t}{n+1} = \log_{2t}{1025} \approx \log_{2t}{1024}$$

total time $= f(t) = (1000+64t)(\log_{2t} 1024) = \ln 1024\left(\frac{1000+64t}{\ln 2t}\right)$

$$f'(t) = \ln 1024\left(\frac{64\ln(2t) - \frac{1000+64t}{t}}{\ln(2t)}\right)$$

$$= \ln 1024\left(\frac{64t\ln(2t) - 64t - 1000}{t\ln(2t)}\right)$$

```
t = 2, time = 5640.0
t = 3, time = 4611.29
t = 4, time = 4186.67
t = 5, time = 3973.6
t = 6, time = 3860.57
t = 7, time = 3803.17
t = 8, time = 3780.0
t = 9, time = 3779.44
t = 10, time = 3794.6
t = 11, time = 3821.11
t = 12, time = 3856.08
t = 13, time = 3897.51
t = 14, time = 3943.96
t = 15, time = 3994.38
t = 16, time = 4048.0
t = 17, time = 4104.21
t = 18, time = 4162.54
t = 19, time = 4222.62
```

let $y(t) = 64t\ln(2t) - 64t - 1000$

$\because y'(t) = 64(\ln(2t)) > 0 \ \forall t > 0$

$f''(t) > 0 \ \forall t > 0$

also $t \in N$

$\Rightarrow t = 9$ minimizes the Disk-Read time.

2-2

$$A[u][v] = \begin{cases} 1 & \text{if } (u,v) \in E \\ 0 & \text{otherwise} \end{cases}$$

when $n=1$

$$A^n[u][v] = A[u][v]$$

when $n=k$

$A^k[u][v]$ = the number of paths of length k from u to v

when $n=k+1$

$$A^{k+1}[u][v] = (A^k A)[u][v] = \sum_{w \in V} A^k[u][w]\, A[w][v],$$

where $A^k[u][w]$ is the number of paths of length, w, which is the previous node of v if $A[w][v]=1$.

$Path_{u \to w \to v} = A^k[u][w] \times A[w][v]$, and we will have to sum up the paths from all possible $w \in V$:

$$\text{Total paths} = \sum_{w \in V} (A^k[u][w] \times A[w][v])$$

By the definition of matrix multiplication, the (u,v)-th entry of $A^{k+1} = A^k A$ is the dot-product of the u^{th} row of A^k and the v^{th} column of $A \Rightarrow A^{k+1}[u][v] = \sum_{w \in V} A^k[u][w] \times A[w][v]$

Therefore, we can conclude that the statement is true.